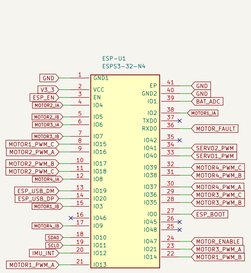
controller / esp32-s3



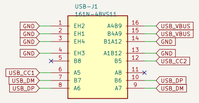


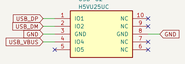




gpio46 is input-only: motor enable defaults low

usb-c / programming and logic power





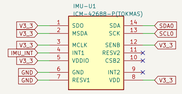


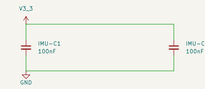






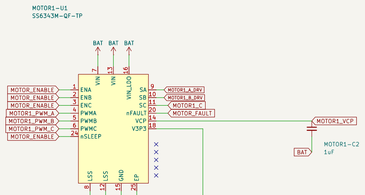
imu / tokmas variant





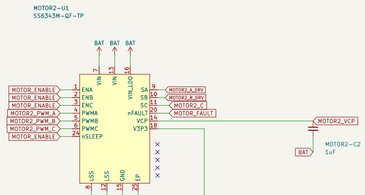
i2c address 0x37: use the tokmas register map

motor 1 / ss6343m



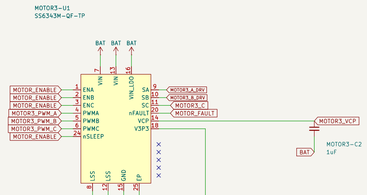


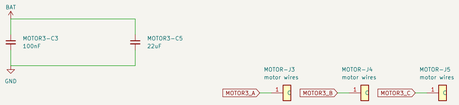
motor 2 / ss6343m



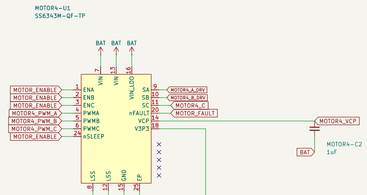


motor 3 / ss6343m





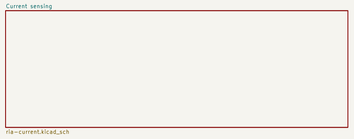
motor 4 / ss6343m



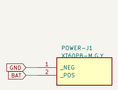


phase current sensing / four motors





power entry, battery sensing and motor control











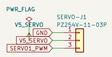


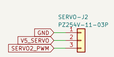






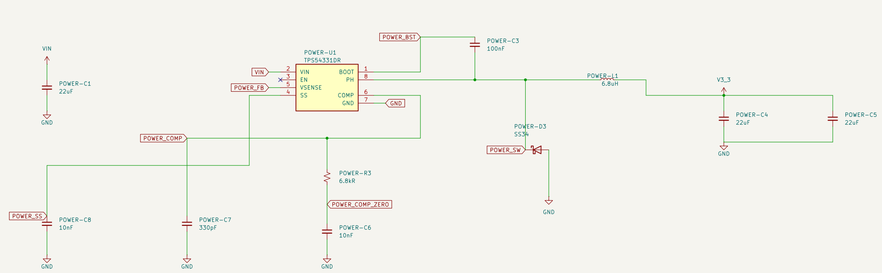




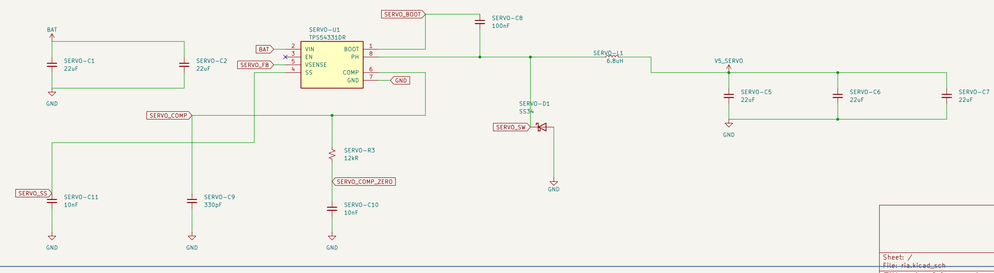


4s li-ion / lipo: 16.8 v maximum charged voltage

logic power / 3.3 v

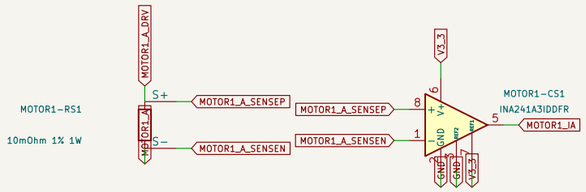


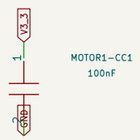
servo power / 5.07 v, 2 a

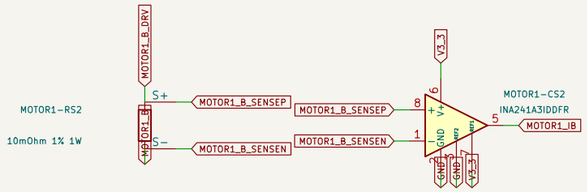


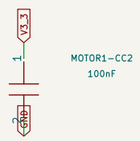
Motor 1 / A and B phase current

10 mOhm x 50 V/V = 0.5 V/A; zero = 1.65 V
Kelvin connections at shunt pads; C phase = −(A + B)



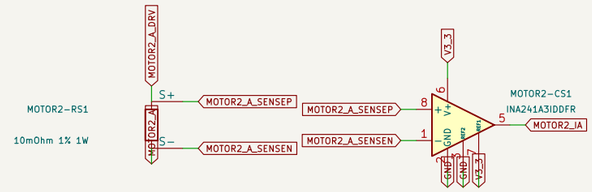


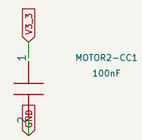


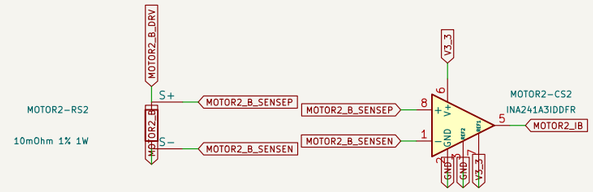


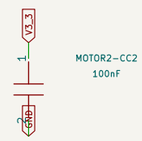
Motor 2 / A and B phase current

10 mOhm x 50 V/V = 0.5 V/A; zero = 1.65 V
Kelvin connections at shunt pads; C phase = −(A + B)



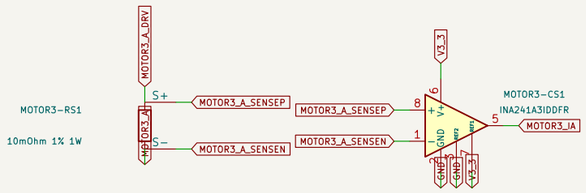


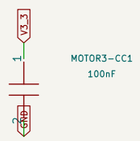


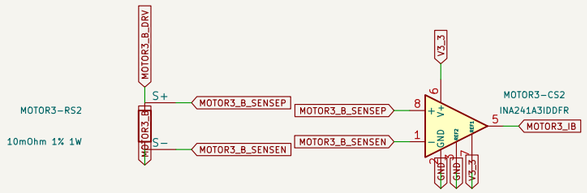


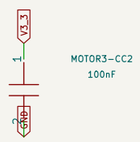
Motor 3 / A and B phase current

10 mOhm x 50 V/V = 0.5 V/A; zero = 1.65 V
Kelvin connections at shunt pads; C phase = −(A + B)



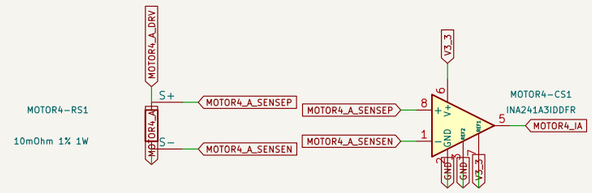


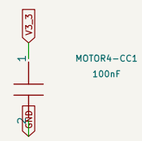


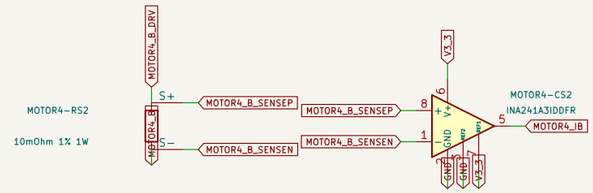


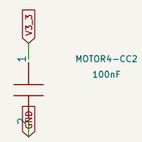
Motor 4 / A and B phase current

10 mOhm x 50 V/V = 0.5 V/A; zero = 1.65 V
Kelvin connections at shunt pads; C phase = −(A + B)









Sheet: /Current sensing/
File: ria−current.kicad_sch

Title: ria / phase current sensing

Size: A3	Date: 2026−11−03	**Rev: four motors**
KiCad E.D.A. 10.99.0−unknown		Id: 2/2